Lecture 2. Trade and Technology: The Ricardian Model

Lectures on International Trade

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Reasons for Trade

Reasons for Trade:

- Difference in technology (i.e. productivity)
- Difference in resources (labor, capital, land, etc.)
- Difference in the costs of offshoring (i.e. various parts of a good are produced in different countries and then assembled)
- Proximity (geographic distance, regional free-trade-agreement)
- Other reasons...

Outline

Trade and Technology:

- A reason why countries trade: different technology
  - Absolute advantage
  - Comparative advantage
- Ricardian model
  - Labor productivity
  - Autarky equilibrium
  - Trade equilibrium

Motivating Facts

Technology and Trade

- Technologies differ across countries.
- Some countries may have superior technology in producing all goods.
- However, they still import a lot from other countries. For example, the United States imports a lot of snowboards from China and Taiwan.

Motivating Facts, cont'

Shaun White's snowboard

- In 2005 the United States imported \$59 million of snowboards from 20 different countries.
- China exported more than \$18 million worth of snowboards to the United States in 2005 and \$21 million in 2009.

Motivating Facts, cont'

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- In 2005 the United States imported \$59 million of snowboards from 20 different countries.
- China exported more than \$18 million worth of snowboards to the United States in 2005 and \$21 million in 2009.
- With all the manufacturing capability in the United States, why does it purchase snowboards from other countries instead of producing them domestically?

U.S. Imports of Snowboards, 2005 and 2009

Rank	Country	Value of Imports 2005 ($ millions)	Value of Imports 2009 ($ millions)	Quantity of Snowboards 2009 (thousands)	Average Price 2009 ($/board)
1	China	$18.15	$21.30	365.0	$58
2	Austria	17.77	11.04	104.0	106
3	Canada	9.11	3.94	51.8	76
4	Mexico	4.96	0.00	0.0	na
5	Spain	2.17	0.57	6.6	85
6	Poland	1.90	0.04	0.5	71
7	Tunisia	1.29	0.15	4.3	34
8	France	1.12	0.07	0.6	127
9	Germany	1.00	0.02	0.1	227
10	Taiwan	0.49	1.14	37.4	31
11	Bulgaria	0.30	0.00	0.0	na
12	Switzerland	0.18	0.08	0.7	113
13-20	All other countries	0.29	0.18	1.49	123
	Total	58.76	38.51	572	67

Technology and Trade

Absolute advantage

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- Germany in chemicals, machine tools, automobiles, steel products, etc.

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- Germany in chemicals, machine tools, automobiles, steel products, etc.
- Absolute advantage is not a good explanation for trade patterns.

Technology and Trade, cont'

Comparative advantage

- A country has comparative advantage in producing those goods that it produces best compared with how well it produces other goods.

Technology and Trade, cont'

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- Canada: 200 snowboards; 1 automobile;
- China: 80 snowboards; 0.5 automobile.

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- United States: 100 snowboards; 2 automobiles;
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- David Ricardo
- England vs. Portugal: cloth and wine
- Ricardo and Mercantilism

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- Note: Two-country, two-good comparison
- Underlying assumptions: Free trade, mobile labor, constant return to scale, no externality.

The Ricardian Model

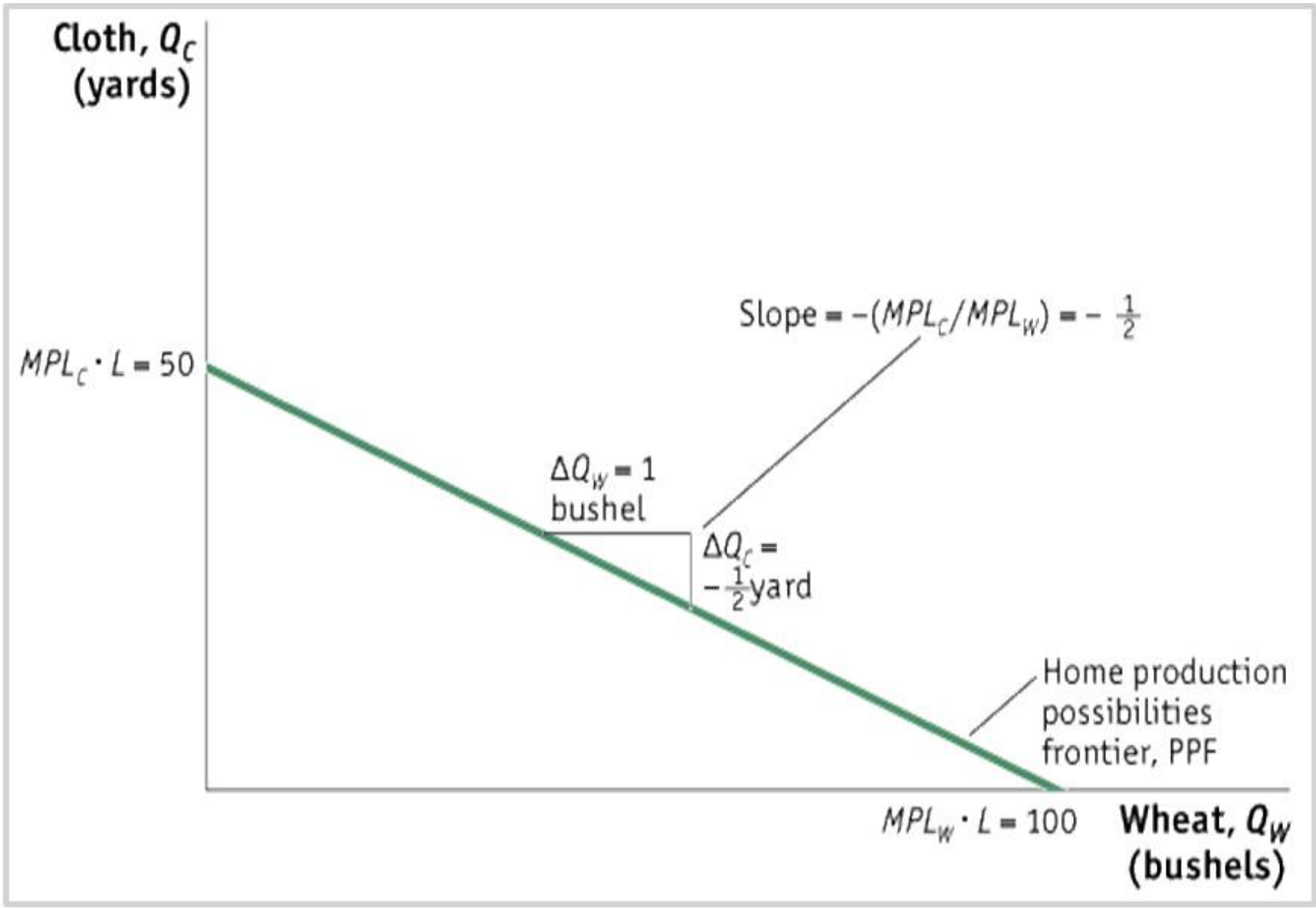
Basic settings:

- One factor of production: labor
- Two goods: wheat and cloth
- Two countries: Home and Foreign

- **MPL**: *marginal product of labor* is the extra output obtained by using one more unit of labor.
- In Home, one worker produces 4 bushels of wheat, so $MPL_W = 4$.
- Alternatively, one worker can produce 2 yards of cloth, so $MPL_C = 2$.

- Assume there are 25 workers at Home.
- **PPF**: *production possibility frontier* is a set of good quantities that a country can produce at maximum.

Home PPF

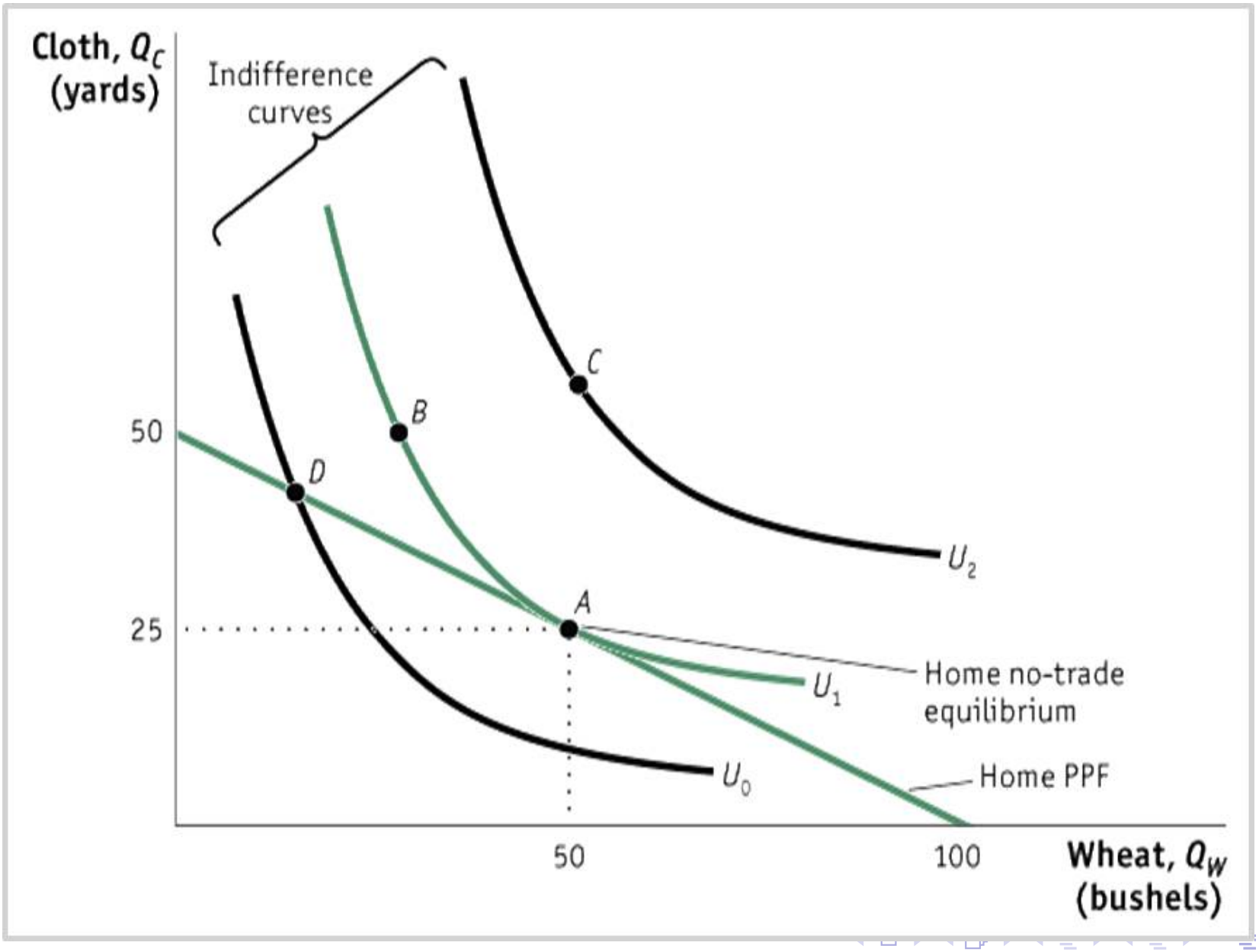

Cloth, Q_C (yards)
Slope = $-(MPL_C/MPL_W) = -\frac{1}{2}$
$MPL_C \cdot L = 50$
$\Delta Q_W = 1$ bushel
$\Delta Q_C = -\frac{1}{2}$ yard
Home production possibilities frontier, PPF
$MPL_W \cdot L = 100$
Wheat, Q_W (bushels)

The Ricardian Model, cont'

Indifference Curve

- A way to represent goods demand.
- Indifference curves are often used to show the preferences of an individual.
- Each indifference curve shows the combinations of two goods, such as wheat and cloth, that a person or economy can consume and be equally satisfied.
- All points on an indifference curve have the same level of utility.
- Points on higher indifference curves have higher utility.

Home Autarky



The Ricardian Model, cont'

Autarky equilibrium

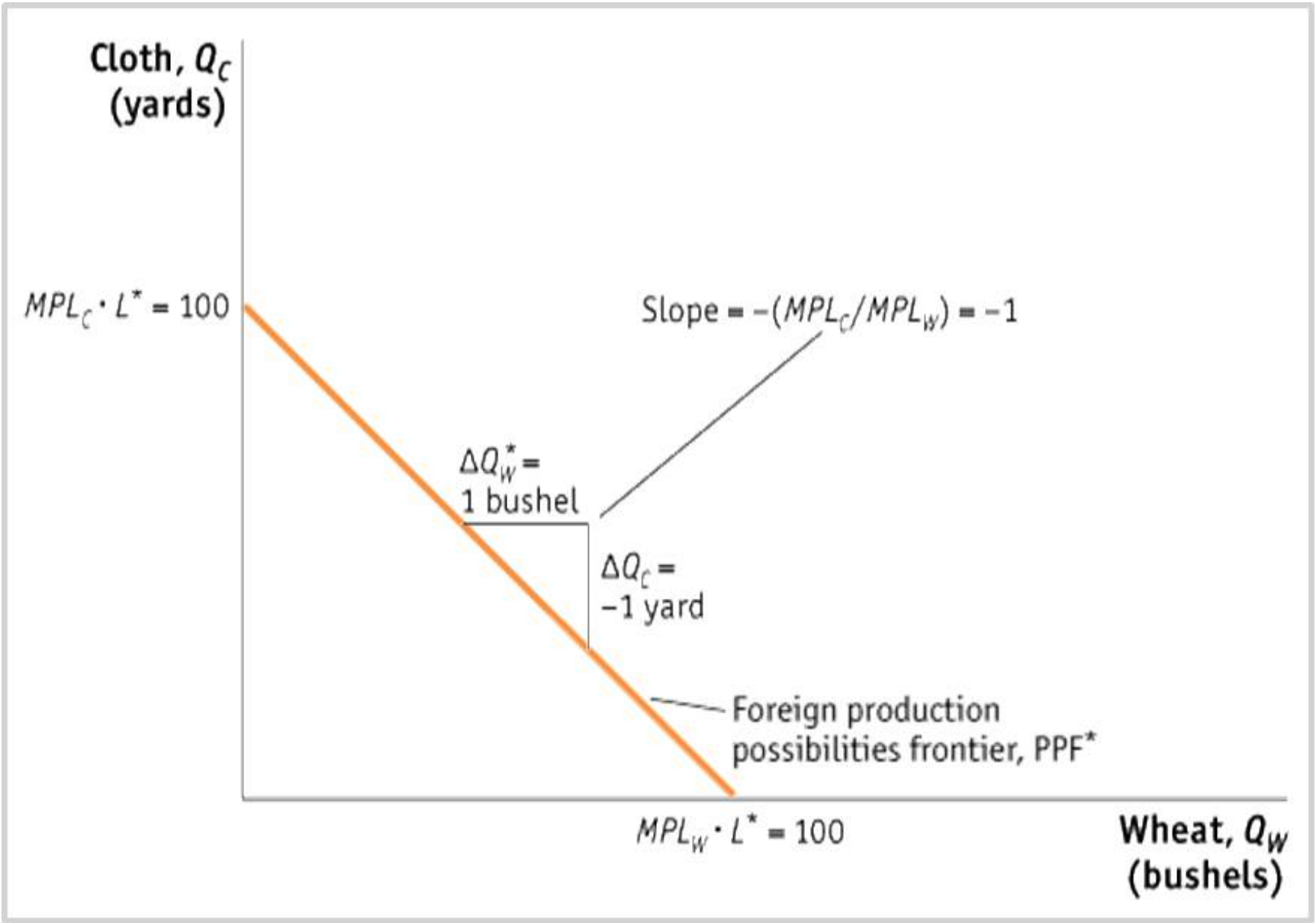
- Opportunity cost and prices: price reflects the opportunity cost of a good under perfect competition.
- Wages
  - In competitive markets firms hire workers up to the point at which the hourly wage equals the value of one more hour of production.
  - The value of one more hour of labor equals the amount of goods produced in that hour (MPL) times the price of the good.
  - Labor hired up to the point where wage equals $P \cdot MPL$ for each industry.
- Wages should equal across industries: labor is mobile
  - $P_W \cdot MPL_W = P_C \cdot MPL_C$
  - $\frac{P_W}{P_C} = \frac{MPL_C}{MPL_W}$

The Ricardian Model, cont'

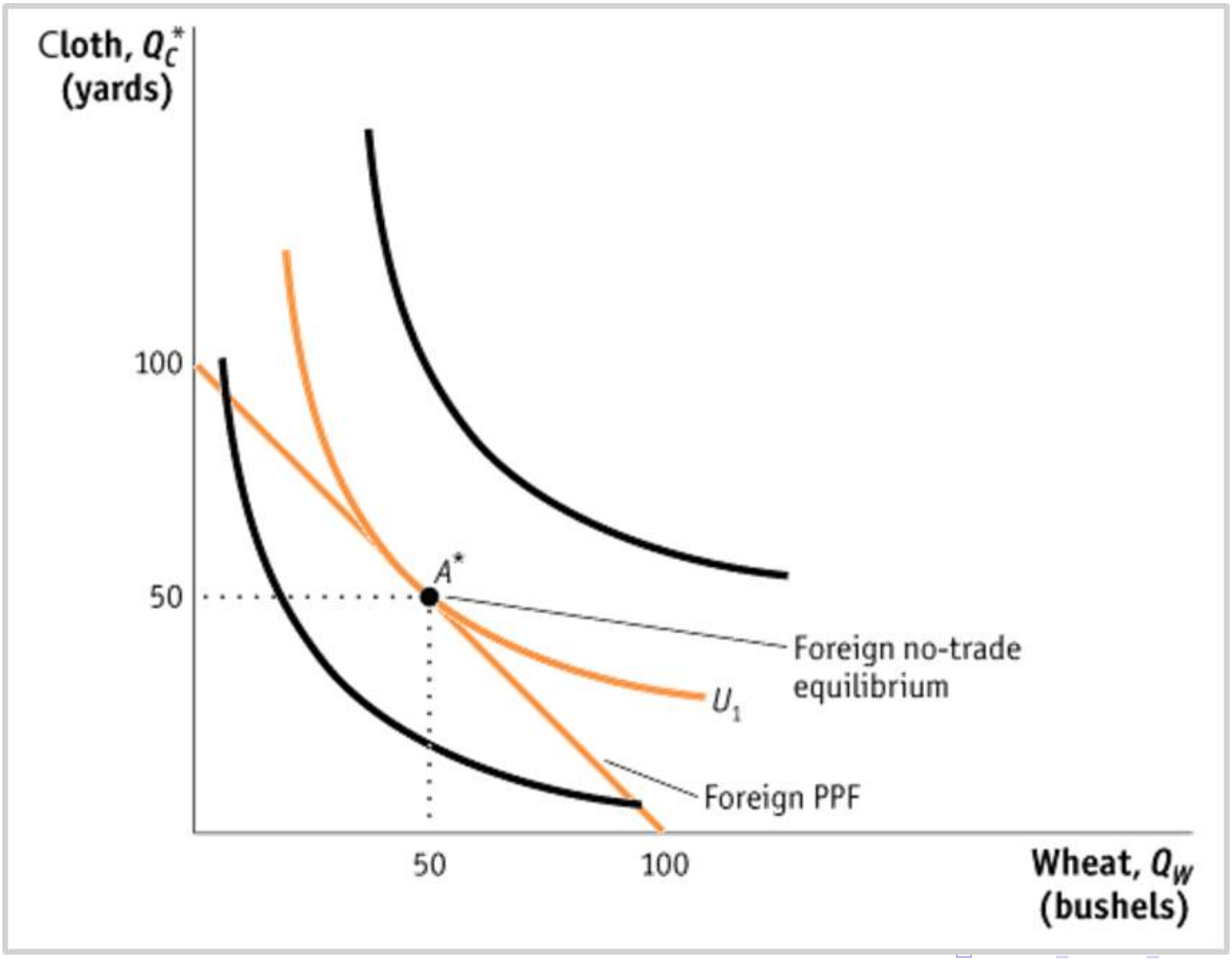
The Foreign Country

- Assume a Foreign worker can produce one bushel of wheat or one yard of cloth.
- $MPL_W^* = 1$; $MPL_C^* = 1$.
- Assume there are 100 workers available in Foreign.
- The relative good price is one.

Foreign PPF


Cloth, Q_C (yards)
$MPL_C \cdot L^* = 100$
Slope = $-(MPL_C/MPL_W) = -1$
$\Delta Q_W^* =$ 1 bushel
$\Delta Q_C =$ −1 yard
Foreign production possibilities frontier, PPF*
$MPL_W \cdot L^* = 100$
Wheat, Q_W (bushels)

Foreign Autarky



Comparative advantage for Home and Foreign

	Cloth (1 Yard)	Wheat (1 Bushel)
Home	2 Bushels of Wheat	½ Yard of Cloth
Foreign	1 Bushel of Wheat	1 Yard of Cloth

- A country has a comparative advantage in a good when it has a lower opportunity cost of producing than another country.
- Foreign has a comparative advantage in producing cloth. Home has a comparative advantage in producing wheat.

Application in Real World

TABLE 2-2 Apparel, Textiles, and Wheat in the United States and China

Sales per employee for the apparel and textile industries in the United States and China, as well as bushels per hour in producing wheat.

The United States has an absolute advantage in all of these products, but it has a comparative advantage in producing wheat.

	United States	China	Absolute Advantage
	Sales/Employee	*Sales/Employee*	*U.S./China Ratio*
Apparel	$92,000	$13,500	7
Textiles	$140,000	$9,000	16
	Bushels/Hour	*Bushels/Hour*	*U.S./China Ratio*
Wheat	27.5	0.1	275
	Comparative Advantage		
Wheat/apparel ratio × 1,000	0.3	0.01	
Wheat/textile ratio × 1,000	0.2	0.01	

The Ricardian Model, cont'

International Trade Equilibrium

- What happens when goods are traded between Home and Foreign?
- Country's no-trade relative price determines which product it will export and which it will import.
- The no-trade relative price equals its opportunity cost of production.
- The pattern of exports and imports will be determined by the opportunity costs of production in each country, i.e. their comparative advantage.

The Ricardian Model, cont'

International Trade Equilibrium

- The relative price of cloth in Foreign is $P_C/P_W = 1$.
- The relative price of cloth in Home is $P_C/P_W = 2$.
- Therefore Foreign would want to export their cloth to Home, since they can make \$1 and export it for more than \$1.
- The opposite is true for wheat.
- Home will export wheat and Foreign will export cloth.
- Both countries export the good for which they have the comparative advantage.

The Ricardian Model, cont'

International Trade Equilibrium

- The two countries are in an international trade equilibrium when the relative price of wheat/cloth is the same in the two countries.
- Trade equilibrium
  - International good prices and wages
  - Production in two countries from autarky to trade
  - Consumption in two countries from autarky to trade

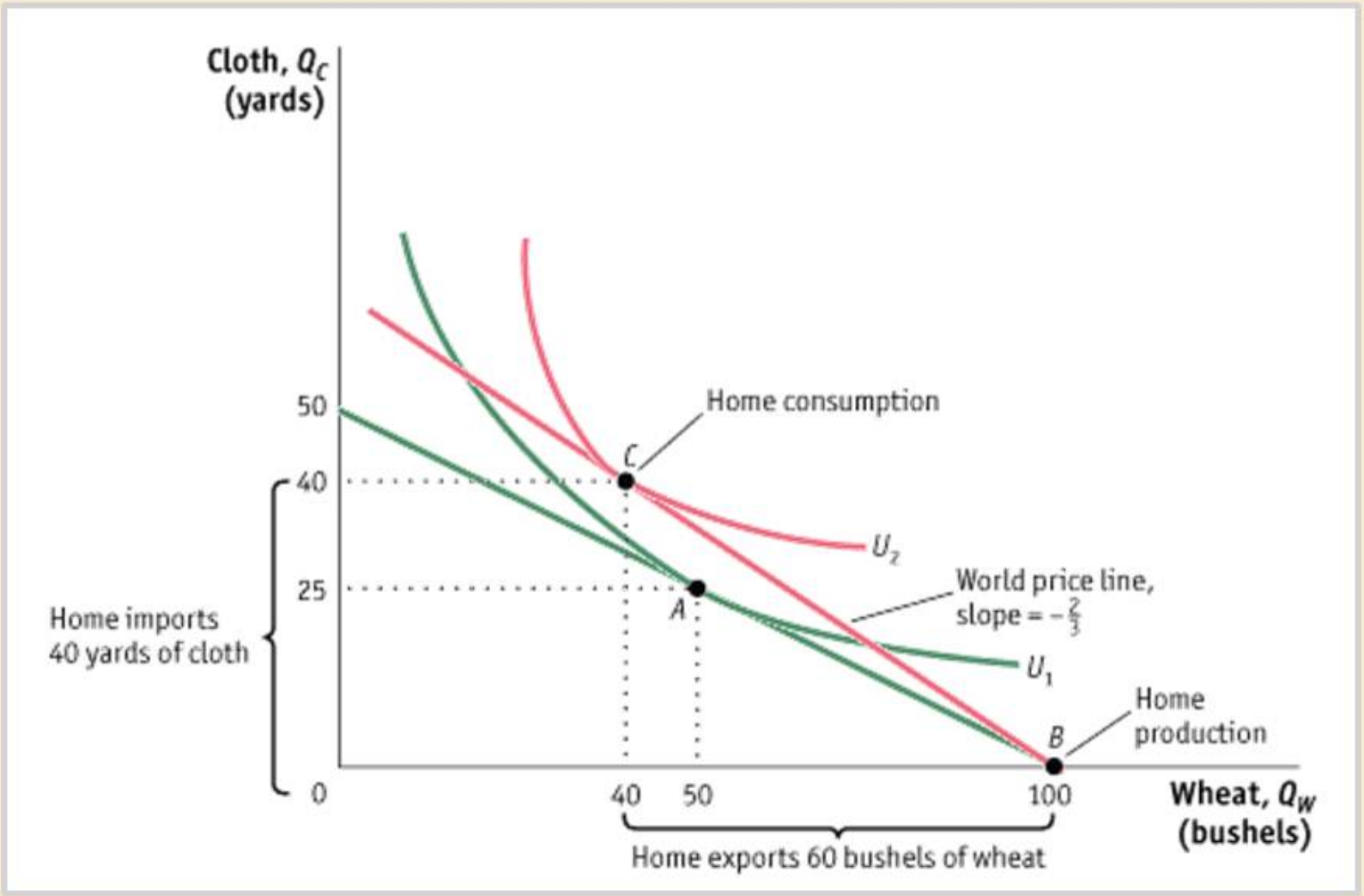
The Ricardian Model, cont'

International Good Price

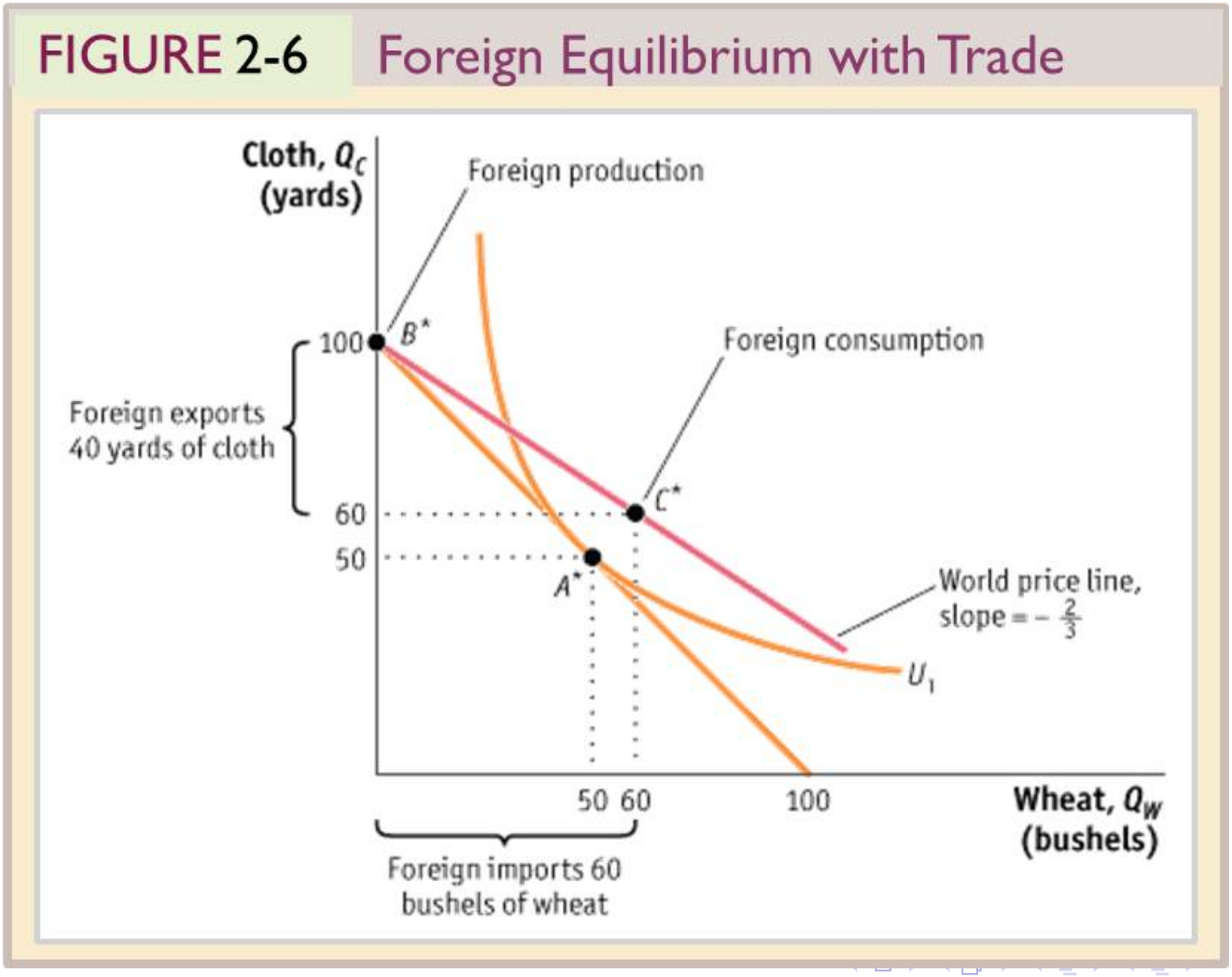
- The relative price of wheat in the trade equilibrium will be between the no-trade price in the two countries.
- For now we will assume the free-trade price of P_W/P_C is $2/3$. This is between the price of $1/2$ in Home and 1 in Foreign.
- We can now take this price and see how trade changes production and consumption in each country.
- The world price line shows the range of consumption possibilities that a country can achieve by specializing in one good and engaging in international trade.

Home Trade Equilibrium

FIGURE 2-5 Home Equilibrium with Trade



Foreign Trade Equilibrium



The Ricardian Model, cont'

Pattern of Trade and Gains from Trade

- The first lesson of the Ricardian model.
  - Each country is exporting the good for which it has the comparative advantage.
  - This confirms that the pattern of trade is determined by comparative advantage.

The Ricardian Model, cont'

Pattern of Trade and Gains from Trade

- The first lesson of the Ricardian model.
  - Each country is exporting the good for which it has the comparative advantage.
  - This confirms that the pattern of trade is determined by comparative advantage.
- The second lesson of the Ricardian model.
  - There are gains from trade for both countries.

Wages Across Countries with Trade

- Real wage at Home
  - Nominal wage: $W = MPL_W \cdot P_W^T$
  - $MPL_W \cdot P_W^T / P_W^T = 4$ bushels of wheat
  - $MPL_W \cdot P_W^T / P_C^T = \frac{8}{3}$ yards of cloth
- Real wage in Foreign
  - Nominal wage: $W^* = MPL_C^* \cdot P_C^T$
  - $MPL_C^* \cdot P_C^T / P_C^T = 1$ yard of cloth
  - $MPL_C^* \cdot P_C^T / P_W^T = \frac{3}{2}$ bushels of wheat
- Foreign workers earn less than Home workers as measured by their ability to purchase either good.
- This fact reflects Home's absolute advantage in the production of both goods.

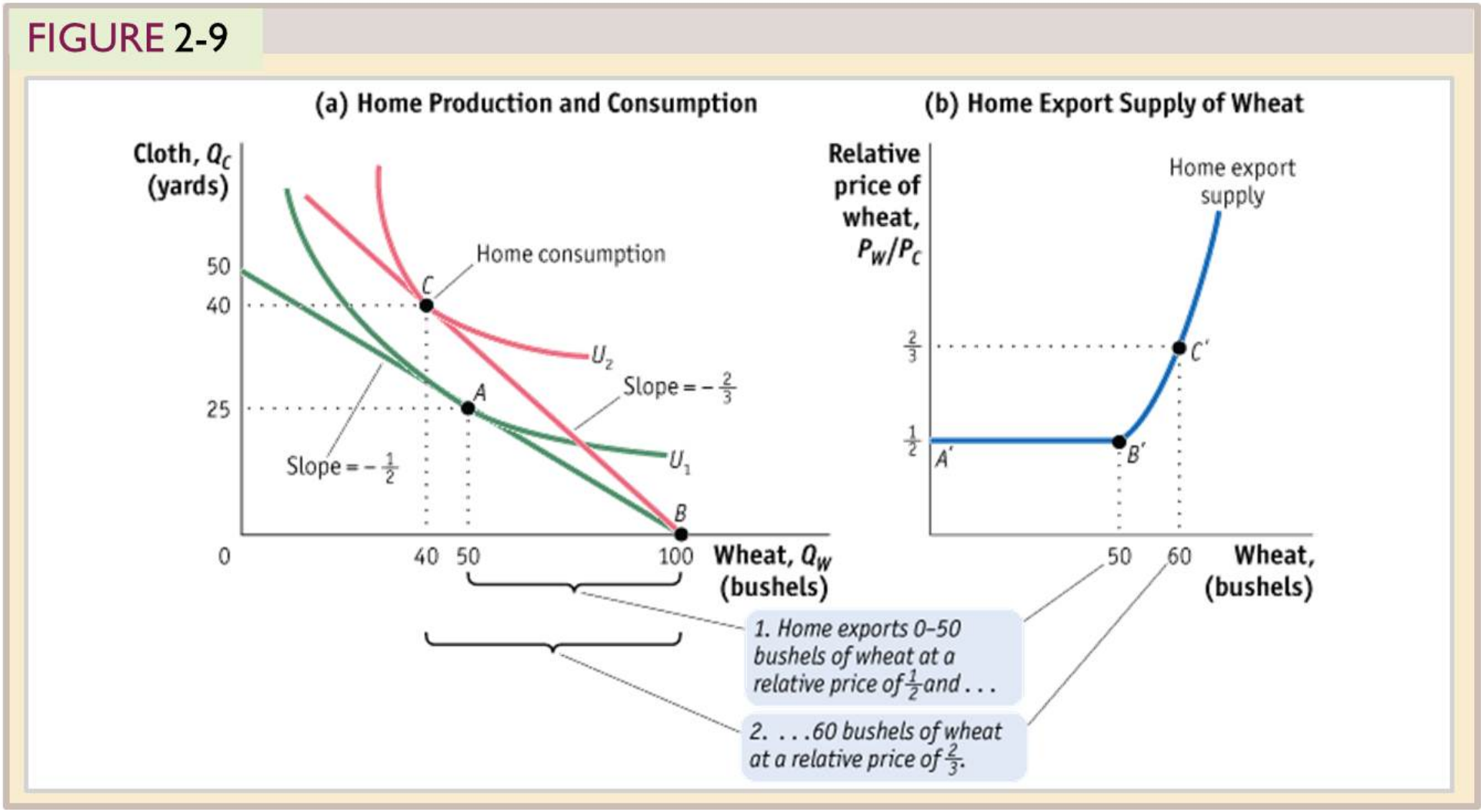
Solving for International Prices

Consider the world market for wheat:

- Home exports wheat, so we will derive a Home export supply curve, which shows the amount it wants to export at various relative prices.
- Foreign imports wheat, so we will derive a Foreign import demand curve, which shows the amount of wheat that it will import at various relative prices.
- Note: One can also consider the world market for cloth. Same results will arise.

Export Supply of Wheat

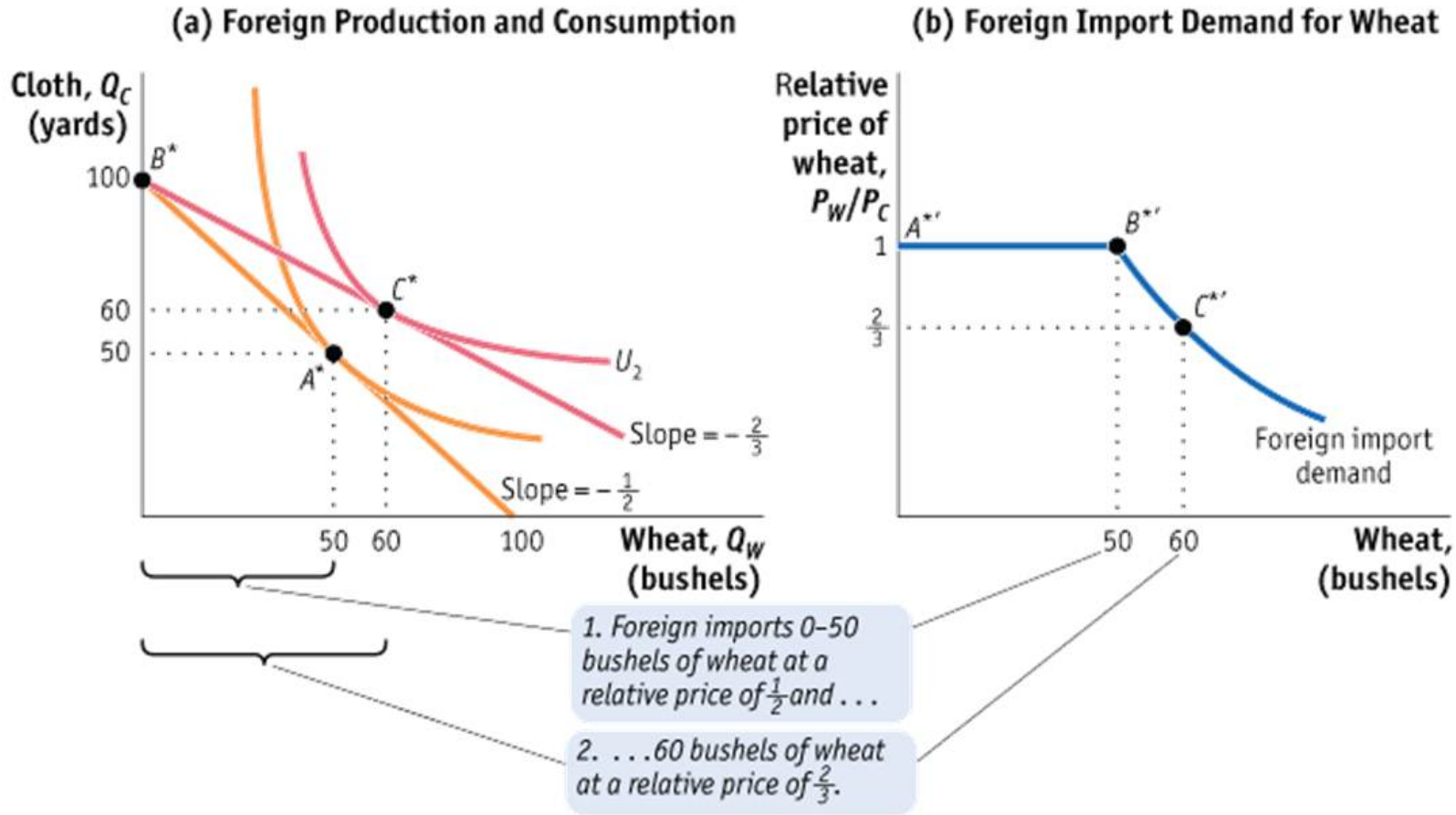
Export Supply from Home:



Import Demand for Wheat

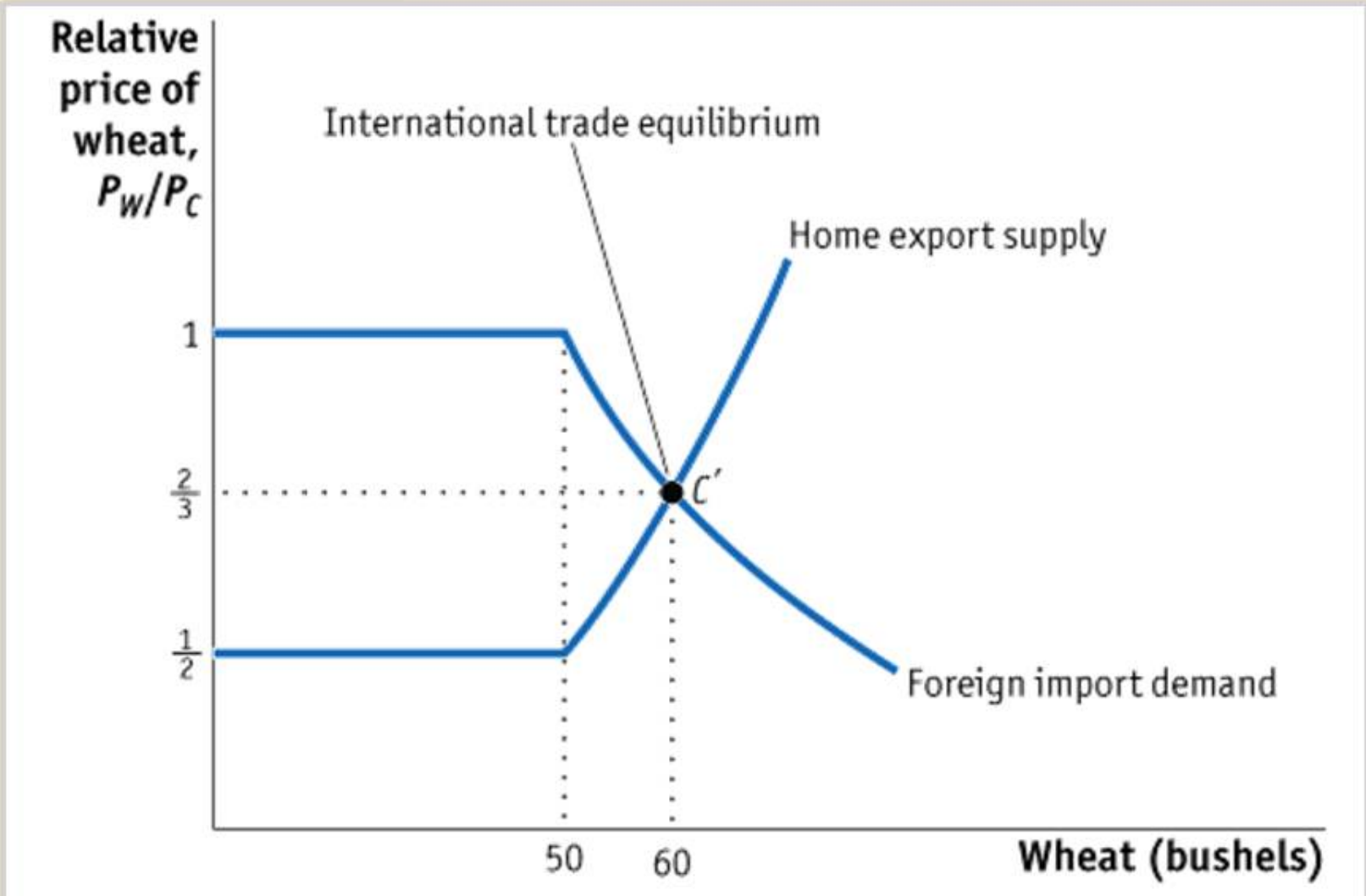
Import Demand From Foreign:

FIGURE 2-10



World Market Equilibrium for Wheat

FIGURE 2-11



World Market Equilibrium Price, An Example/Exercise

- In the previous setting of Home and Foreign countries, in the open world economy, try to draw the relative supply of wheat (relative to supply of cloth) against the relative price of wheat (relative to price of cloth), with $\frac{P_w}{P_c}$ on the Y-axis and $\frac{Q_w}{Q_c}$ on the X-axis.
- Assume utility function given by

$$U = Q_w^{0.4} Q_c^{0.6},$$

  Draw the relationship between relative price and relative demand for wheat.
- Determine the relative price for wheat in the open economy equilibrium.

The Ricardian Model, summary

The Ricardian Model

- Basic settings
- Autarky equilibrium
- Trade pattern: comparative advantage
- Trade equilibrium
  - International good prices
  - Wages across countries
  - Production and consumption with trade
- Gains from trade

Summary

Key Points:

1. A country has comparative advantage in producing a good when the country's opportunity cost of producing the good is lower than the opportunity cost of producing the good in another country.

2. The pattern of trade between countries is determined by comparative advantage. Even countries with poor technologies can export the goods in which they have comparative advantage.

3. All countries experience gains from trade. That is, the utility of an importing or exporting country is at least as high as it would be in the absence of international trade.

Summary, cont'

4. The level of wages in each country is determined by its absolute advantage, i.e. the amount one can produce with its labor. This result explains why countries with poor technologies are still able to export: their low wages allow them to overcome their low productivity.

5. The equilibrium price of a good on the world market is determined at the point where the export supply of one country equals the import demand of the other country.

6. A country's **Terms of Trade** equal the price of its export good divided by the price of its import good. A rise in a country's terms of trade makes it better off because it is exporting at higher prices or importing at lower prices.

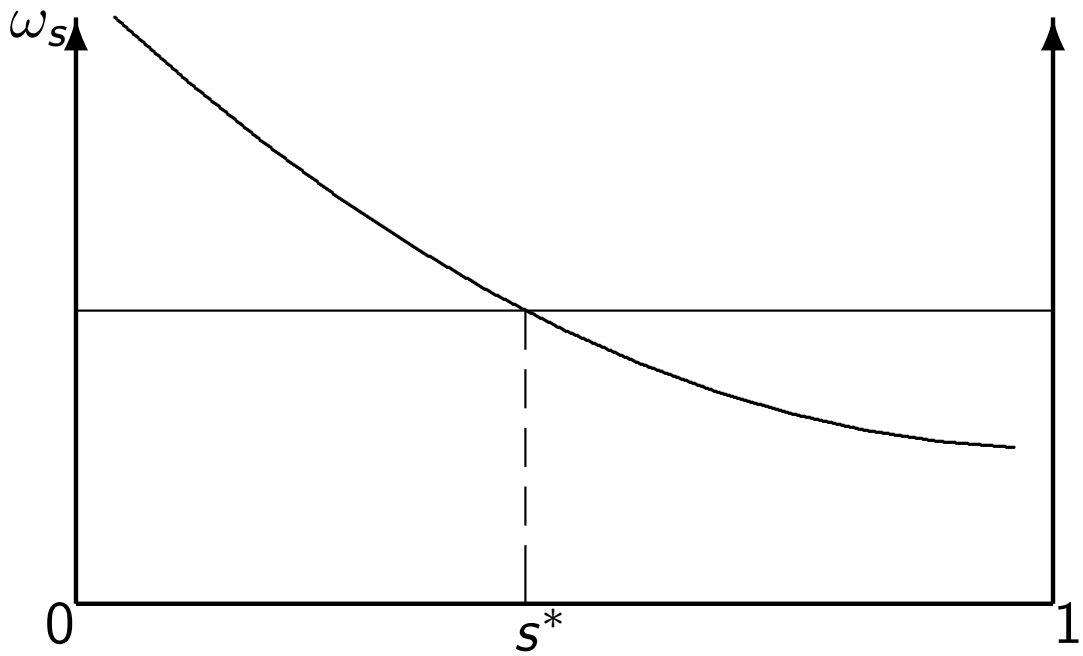
An Extension of the Ricardian Model

Reference: Dornbusch, R., S. Fischer and P. A. Samuelson. 1977. "Comparative Advantage, Trade, and Payments in a Ricardian Model with a Continuum of Goods." The American Economic Review, Vol. 67, No. 5, pp. 823-839.

- A continuum of industries, with productivity a_s, $s \in [0, 1]$
- Two countries: H and F
- Relative productivity for each industry: $\frac{a_s^H}{a_s^F} \equiv \omega_s$
- Rank industries by ω_s
- Comparative advantage in relatively more productive industries

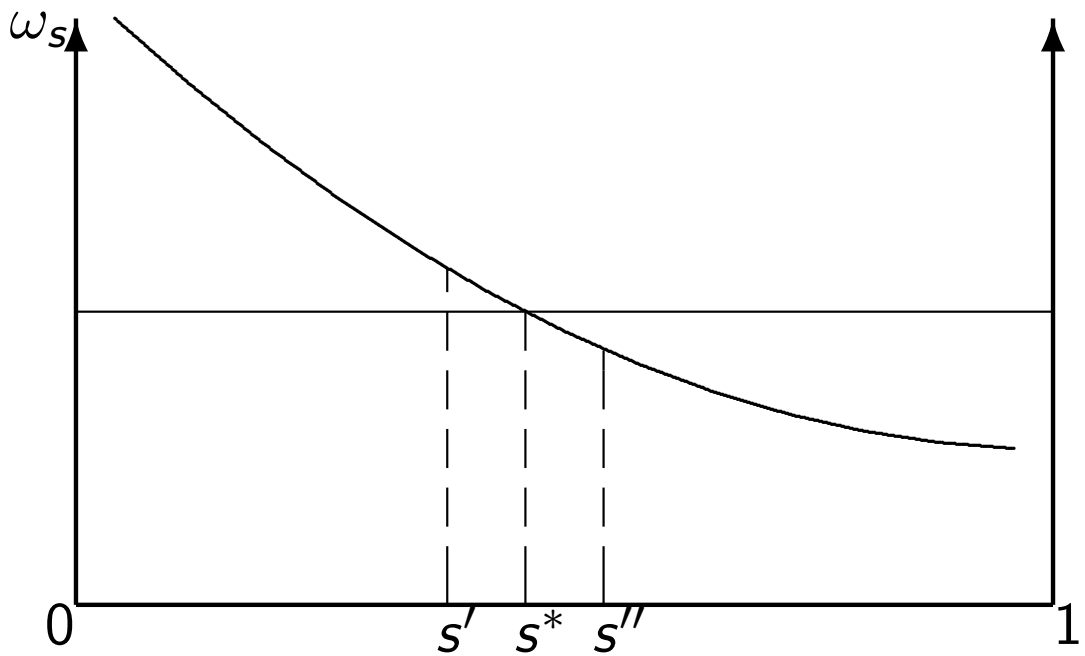
An Extension of the Ricardian Model (cont.)

Comparative advantage across a continuum of industries: cutoff industry is determined in equilibrium by trade balance condition.



An Extension of the Ricardian Model (cont.)

Question: What if there is proportional iceberg cost τ for international trade?



Two cutoffs: $\omega_{s'} = (1+\tau)\omega_{s''}$.